

Crowded Carry: Crash Risk in FX and Crypto Carry Trades

MFE 230GB Currency Markets, final project

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GitHub repository: <https://github.com/rezazamani2329/currency-crypto-carry>

Presentation (10 minutes): [slides/presentation_10min.pptx](#) · [PDF](#) · [slides with speaker notes \(PDF\)](#) · speaker notes: [md](#) · [PDF](#)

Other material: interactive results page [web/index.html](#) · long deck [slides/presentation.pptx](#) · step-by-step notebooks [notebooks/](#) · study guide [docs/study-guide.html](#) ([PDF](#), Persian [HTML](#) / [PDF](#)) · all online links [docs/LINKS.md](#)

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0. Summary: question and answer

Our question: do carry trades crash when they are crowded?

Short answer: partly yes. Crowding makes carry weaker, but it doesn't predict the biggest crashes.

- **FX:** carry crashes are compensation for risk, and crowding is a modest warning sign, not a reliable crash predictor. After crowded months, next-month carry averages 0.02% against 0.30% otherwise, and halving positions when crowded lifts the out-of-sample Sharpe ratio from 0.27 to 0.32. But the 2008 crash started from a non-crowded reading, so the maximum drawdown is -37% with or without the filter.
- **Crypto:** carry also earns a premium and also crashes, but the crashes come from single coins, mostly being long a coin that traders are crowding to short, so crowding tells you where the danger is rather than when it will hit. Funding pays about 8.5% a year, but price moves decide the result (net Sharpe 0.14), and the worst week was the Terra (LUNA) collapse, -27%.
- **Together:** the two carry trades are almost uncorrelated (0.11), so a 50/50 mix has lower volatility and a smaller drawdown than either one alone.

Headline numbers, net of costs	Net return a year	Volatility a year	Sharpe	Max drawdown
A: G10 carry, 1999–2026	3.3%	8.9%	0.37	–37%
A: crowd-filtered carry, 1999–2026	3.2%	8.5%	0.38	–37%
C: crypto funding carry, 2020–2026	4.1%	30%	0.14	–67%
C: crypto funding carry, since 2024	13.9%	17%	0.81	–13%

1. Motivation

A carry trade buys high-yielding assets and funds them with low-yielding ones. Uncovered interest parity (UIP) says this should earn nothing, because the high-yield currency should fall by exactly the interest gap. In the data it earns a premium instead (the forward premium puzzle). The usual explanation is crash risk: carry goes **"up the stairs and down the elevator"**, with small steady gains most of the time and then large losses in a few weeks. Crashes are worst when the trade is crowded and everyone unwinds at once (Brunnermeier, Nagel & Pedersen, 2008).

We test this in two markets with the same economics but very different investors:

	Strategy A: crowded G10 carry	Strategy C: crypto funding carry
Market	Developed-market currencies	Crypto perpetual futures (Binance)
Universe	Japanese yen (JPY), euro (EUR), British pound (GBP), Swiss franc (CHF), Canadian dollar (CAD), Australian dollar (AUD), New Zealand dollar (NZD), each against the US dollar	Bitcoin (BTC), Ethereum (ETH), BNB, XRP, Cardano (ADA), Dogecoin (DOGE), Solana (SOL), Litecoin (LTC), Chainlink (LINK), Avalanche (AVAX), Terra (LUNA), FTX Token (FTT)
Carry signal	3-month interest rate minus the US rate	Perpetual-futures funding rate
Crowding data	CFTC speculator positions	Funding itself measures leveraged demand

	Strategy A: crowded G10 carry	Strategy C: crypto funding carry
Rebalancing	Monthly	Weekly (Sunday), daily P&L
Sample	1999-05 to 2026-08	2020-02 to 2026-08

2. Research question and sub-questions

Main question: do carry trades crash when they are crowded? We split it into six parts:

1. **RQ1.** Does G10 carry earn a positive risk-adjusted return after costs, and does it reproduce the academic carry factor?
2. **RQ2.** Does crowding, measured from CFTC positioning, predict weaker or more crash-prone carry, and does cutting exposure when crowded improve performance out of sample?
3. **RQ3.** When does FX carry lose money? Is it exposed to volatility shocks and the dollar factor?
4. **RQ4.** Does crypto funding carry (short high-funding coins, long low-funding coins) earn a premium after costs, and does the return come from funding or from price moves?
5. **RQ5.** What drives crypto carry losses?
6. **RQ6.** Are FX carry and crypto carry correlated, and does a combined portfolio diversify?

3. Hypotheses

The hypotheses and all trading parameters were written down **before** any backtest was run.

Strategy A (crowded G10 carry)

- **A1.** H0: G10 carry has zero expected excess return after costs. H1: it earns a positive premium.
- **A2.** H0: next-month carry is the same after crowded and non-crowded months. H1: after crowded months carry is weaker and more negatively skewed, so cutting exposure when crowded raises the out-of-sample Sharpe ratio.
- **A3.** H0: carry is unrelated to volatility shocks. H1: carry loses when volatility jumps.

Strategy C (crypto funding carry)

- **C1.** H0: short high-funding / long low-funding has zero expected return after costs. H1: it earns a positive premium, mainly from the funding it collects.
- **C2.** H0: its returns are driven by the crypto market. H1: being dollar-neutral, it has little BTC exposure, and its crashes are coin-specific.

Additional test: FX and crypto carry diversify each other.

4. Data

All data are public and were downloaded on 2026-10-02.

Data	Source	Series / coverage	Used for
FX spot rates	FRED (Fed H.10), daily → month-end	DEXJPUS, DEXUSEU, DEXUSUK, DEXSZUS, DEXCAUS, DEXUSAL, DEXUSNZ	Strategy A returns
3-month interbank rates	OECD via FRED, monthly	IR3TIB01xxM156N (US, JP, EZ, UK, CH, CA, AU, NZ, DE)	Strategy A carry signal
Speculator positions	CFTC Commitments of Traders, legacy futures-only, weekly	CME currency futures, 1986–2026	Strategy A crowding signal
Crypto price, volume, funding	Binance public archive (data.binance.vision), USDT-margined perpetuals	Daily candles and 8-hourly funding, 2020-01 to 2026-08	Strategy C
Carry factor portfolios	Lustig, Roussanov & Verdelhan, <i>CurrencyPortfolios.xls</i> (course file)	Developed-country HML, dollar factor RX and equity-volatility innovations, 1983–2021	Validation and risk regressions

Coverage of the cleaned data in data/clean/:

- **FX spot:** JPY, GBP, CHF, CAD, AUD, NZD from 1971-01; EUR from 1999-01. All to 2026-09.
- **3-month rates:** long histories for USD, EUR/DEM, GBP, CAD, AUD and NZD, but JPY only from 2002-04 and CHF from 1999-07; GBP ends 2026-01. Strategy A therefore starts in 1999-05.
- **CFTC positions:** CAD, CHF, EUR (Deutsche mark before 1999), GBP and JPY from 1986-01, AUD from 1987-01, NZD from 1999-01 (no NZD contract from 2000-04 to 2003-11).
- **Crypto:** the 12 coins above, plus USDC as a stablecoin sanity check (not traded). Average annualised funding is about 12–15% on the main coins (BTC 11.8%, ETH 13.9%, XRP 14.7%) and positive on 81–88% of days.
- **Data fix:** before August 2000 the CFTC lists currency futures under the exchange name "INTERNATIONAL MONETARY MARKET". Matching both names extends positioning back to 1986.

5. Approach: how the two strategies trade

Each step uses only data available at the time of the trade.

Strategy A: G10 FX carry, monthly

1. **Month-end: rank.** Compute each currency's 3-month interest rate minus the US rate.
2. **Trade.** Buy the 2 highest-rate currencies and sell the 2 lowest, 50% of capital in each leg, so there is no net dollar bet.
3. **Crowding check.** For each currency take speculators' net futures position from the CFTC report. The report shows Tuesday positions and is published on Friday, so it is used only from Friday. Z-

score it over 36 months. Portfolio crowding is the average z of the currencies we buy minus the average z of those we sell. If crowding is above 1, halve all positions.

4. **Hold one month** and earn the currency move plus the interest-rate gap.

5. **Rebalance** at the next month-end: rank again and swap, paying 3 bp per unit traded.

In practice the portfolio is almost always long the New Zealand dollar (94% of months) and the Australian dollar (75%), and short the Swiss franc (99%), often with the yen (52%) or the euro (40%). 33 of 328 months were crowded.

Strategy C: crypto funding carry, weekly

1. **Sunday close: rank.** Average each coin's daily funding rate over the past 7 days, using data up to the day before. A coin counts only on days it has a price and positive volume.

2. **Trade.** Short the third of coins with the highest funding and buy the lowest third, equal weights, 50% of capital in each leg (dollar-neutral). This is usually 3 coins on each side: 2 in early 2020 and briefly 4 in 2022, because the 12 coins were never all trading at once.

3. **Hold 7 days.** Earn the price move plus funding; a short receives funding when it is positive.

4. **Rebalance** next Sunday: rank again and swap, paying 5 bp per unit traded.

There is no separate crowding filter in C, because high funding already is a crowding signal.

6. Model and formulas

Currency excess return (A). With S in USD per foreign currency unit and i the 3-month rate in percent a year:

$$rx(k,t) = (1 + i(k,t-1)/1200) * S(k,t)/S(k,t-1) - (1 + i(US,t-1)/1200)$$

This is the return on a one-month forward position under covered interest parity.

Carry weights (A). At month-end t , rank currencies by $i(k,t) - i(US,t)$; weight +0.5 on each of the top 2 and -0.5 on each of the bottom 2. Portfolio return in month $t+1$: $r(t+1) = \sum w(k,t) \cdot rx(k,t+1)$.

Crowding signal (A).

$pos(k,t) = (\text{non-commercial long} - \text{non-commercial short}) / \text{open interest}$ (last release by month-end)

$z(k,t) = (pos(k,t) - 36\text{-month mean}) / 36\text{-month std}$ (minimum 24 months)

$crowd(t) = \sum w(k,t) \cdot z(k,t) = \text{average } z \text{ of longs} - \text{average } z \text{ of shorts}$

$scale(t) = 0.5 \text{ if } crowd(t) > 1, \text{ else } 1$

Crypto funding carry (C). Signal $f(c,t)$ = trailing 7-day average of daily funding, using data up to day t . Every Sunday, weight $-0.5/k$ on the k highest-funding coins and $+0.5/k$ on the k lowest, with k = one third of tradable coins. Daily P&L per coin:

$$pnl(c,t+1) = w(c,t) * price\ return(c,t+1) - w(c,t) * funding\ paid(c,t+1)$$

A short ($w < 0$) therefore receives positive funding.

Return decomposition (C). Net P&L = funding component + price component – costs.

Transaction costs. $cost(t) = turnover(t) \times c$, with $turnover = \sum |w(t) - w(t-1)|$ and $c = 3$ bp (FX) or 5 bp (crypto).

Risk regressions. Newey-West t-statistics (3 lags monthly, 7 lags daily):

$$\begin{aligned} r_A(t) &= \alpha + \beta_{RX} \cdot RX(t) + \beta_{vol} \cdot \Delta Vol(t) + \varepsilon(t) && \text{monthly, 1999-05 to 2021-05} \\ r_C(t) &= \alpha + \beta_{BTC} \cdot r_{BTC}(t) + \varepsilon(t) && \text{daily} \end{aligned}$$

Performance statistics. Annualised mean, volatility and Sharpe ratio ($\times 12$ for monthly data, $\times 365$ for daily data, square root of these for volatility), maximum drawdown, skewness, worst period, hit rate and turnover.

Combined portfolio. Each strategy is scaled to 10% annual volatility using trailing volatility known at $t-1$ (36 months for A, 12 for C), then combined 50/50.

7. Methodology: keeping the tests honest

- **Pre-specified parameters.** Legs, windows, thresholds, lookbacks, rebalancing and costs were fixed before the backtests and not changed afterwards.
- **In-sample / out-of-sample split.** A: in-sample 1999–2010, out-of-sample 2011–2026. C: in-sample 2020–2023, out-of-sample 2024–2026.
- **Costs.** 3 bp per unit traded for G10 FX and 5 bp for crypto; results are shown gross and net.
- **No look-ahead.** Weights set at the end of t earn the return of $t+1$. CFTC positions (Tuesday) are used only after their Friday release. Crypto signals use funding up to the previous day.
- **No survivorship bias.** Terra (LUNA) and FTX Token (FTT), which collapsed in 2022, stay in the universe. A coin is tradable only on days with a price and positive volume, which removes LUNA after 2022-05 and FTT after 2022-11-14. Only the 5 days missing from Binance's archive are forward-filled.
- **Robustness reported, not used for tuning.** Grids over the crowding threshold and exposure (A) and over lookback $\times$ rebalancing (C) are reported in full; headline results stay pre-specified.
- **External validation.** Strategy A is compared with the Lustig–Roussanov–Verdelhan developed HML carry factor.

8. Results

All numbers are read from the files in [output/](#). Returns are net of costs unless stated.

8.1 Strategy A: G10 carry and the crowding filter (RQ1, RQ2)

Figure 1. Growth of \$1, carry vs crowd-filtered carry (net).

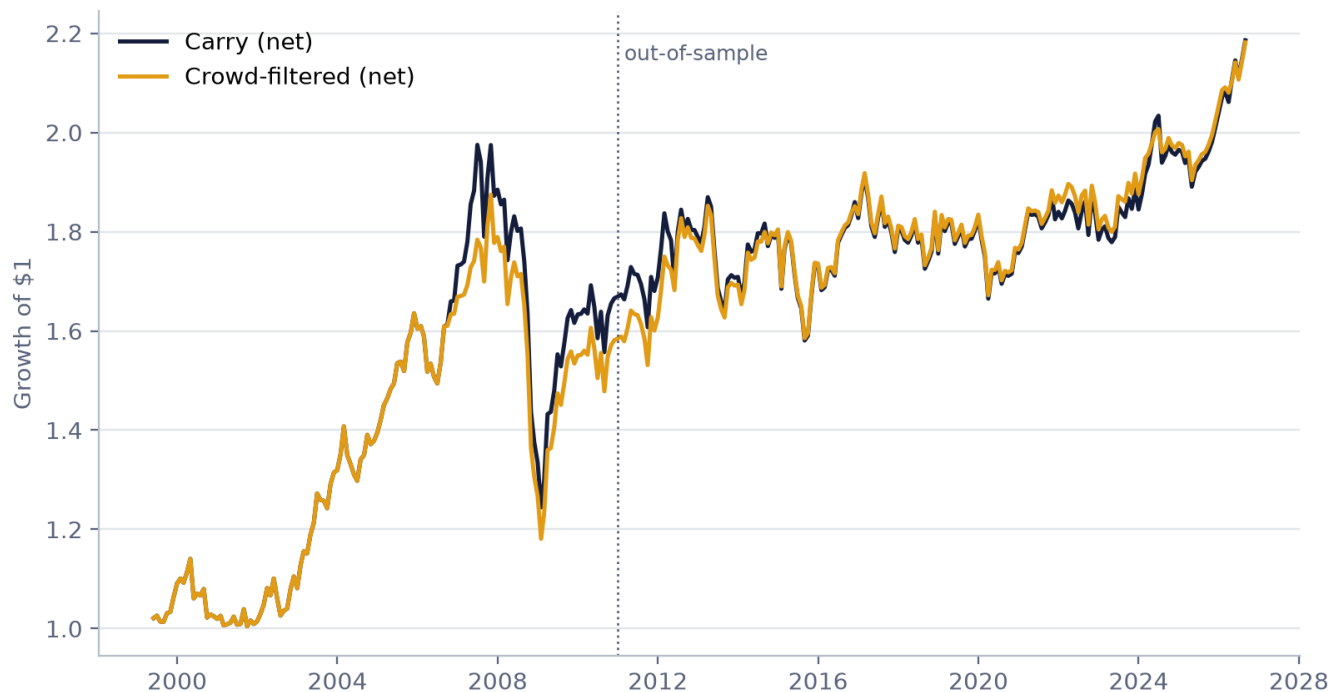


Table 1. Strategy A performance.

Net of costs	Period	Ann. mean	Ann. vol	Sharpe	Max DD	Skew	Turnover
Carry	Full 1999-05 to 2026-08	3.27%	8.91%	0.37	-37.0%	-0.49	1.2x
Carry	In-sample 1999-2010	4.96%	10.55%	0.47	-37.0%	-0.71	1.6x
Carry	Out-of-sample 2011-2026	2.01%	7.46%	0.27	-15.5%	-0.10	0.9x
Crowd-filtered	Full	3.23%	8.55%	0.38	-37.0%	-0.45	2.0x
Crowd-filtered	In-sample	4.47%	10.12%	0.44	-37.0%	-0.69	1.8x
Crowd-filtered	Out-of-sample	2.31%	7.16%	0.32	-14.4%	-0.02	2.1x

Table 2. Carry in the month after crowded vs non-crowded months.

	Months	Next-month mean	Next-month vol	Skewness
Not crowded	295	0.30%	2.56%	-0.46
Crowded	33	0.02%	2.67%	-0.71

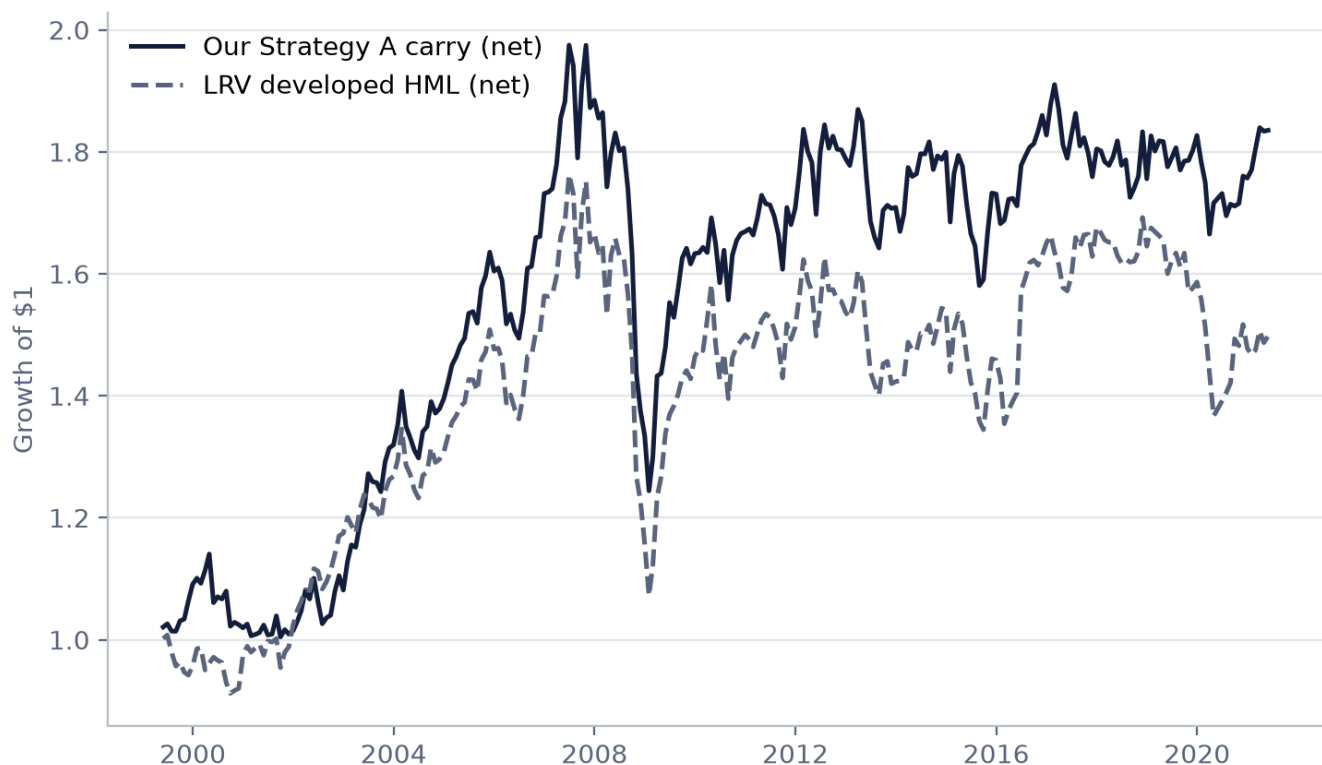
Table 3. Robustness of the crowding filter (net Sharpe).

Threshold	Exposure when crowded	Sharpe full	Sharpe out-of-sample
0.5	0%	0.37	0.31

Threshold	Exposure when crowded	Sharpe full	Sharpe out-of-sample
0.5	50%	0.38	0.30
1.0	0%	0.38	0.37
1.0	50% (pre-set)	0.38	0.32
1.5	0%	0.35	0.39
1.5	50%	0.36	0.33

Every setting beats plain carry out of sample (0.27).

Figure 2. Strategy A vs the academic LRV carry factor.



Validation (RQ1). Over 1999-05 to 2021-05 (264 months), Strategy A has a **0.82 correlation** with the LRV developed HML factor (Strategy A 3.22% a year, Sharpe 0.34; LRV HML 2.34%, Sharpe 0.23).

The full diagnostic chart from the backtest script (cumulative returns, drawdowns and the crowding signal) is output/strategyA/strategyA_performance.png.

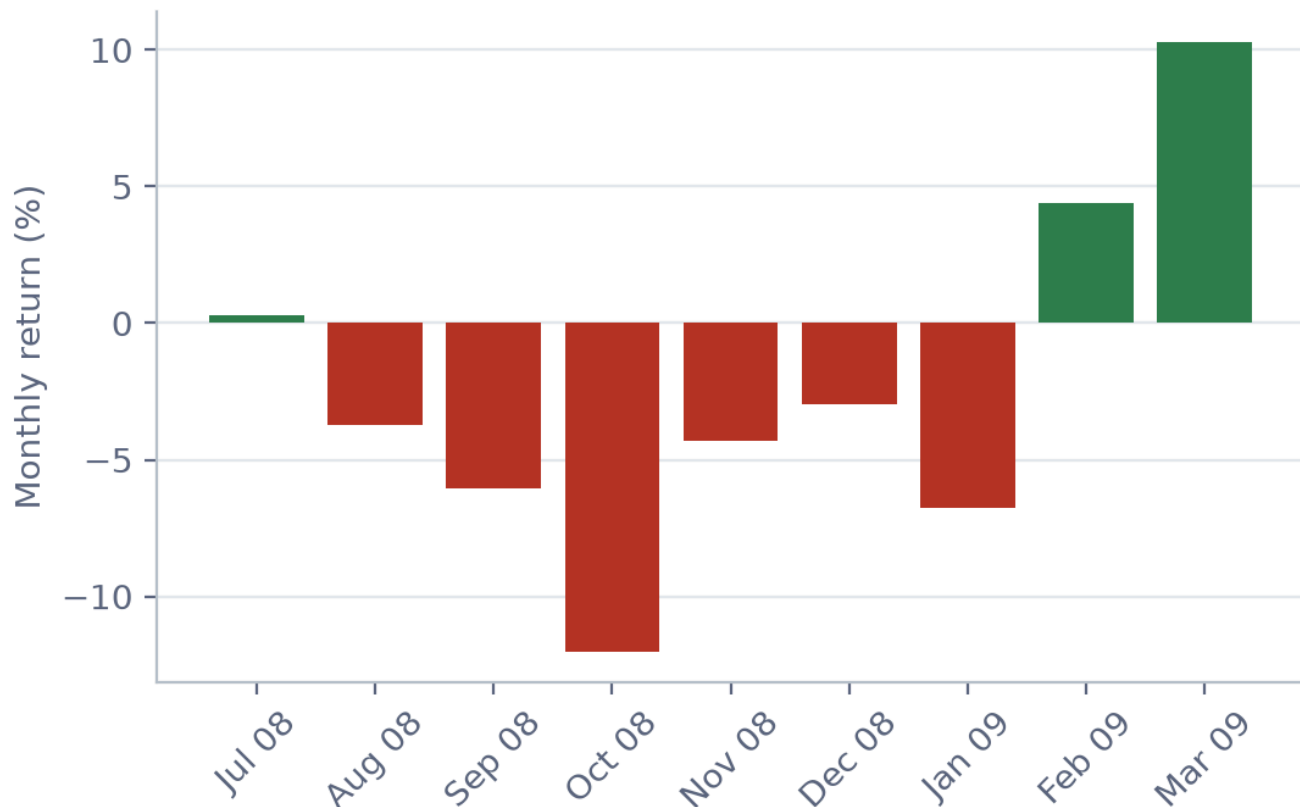
8.2 Strategy A: when does it lose money? (RQ3)

Table 4. Monthly net carry on risk factors, 1999-05 to 2021-05, Newey-West t-statistics.

Model	Dollar factor (RX)	Volatility innovation	Monthly alpha	R ²
RX only	0.42 (t = 4.4)	–	0.26% (t = 1.6)	0.12

Model	Dollar factor (RX)	Volatility innovation	Monthly alpha	R ²
Volatility only	–	-1.93 (t = -4.3)	0.26% (t = 1.6)	0.10
Both	0.35 (t = 4.1)	-1.55 (t = -3.7)	0.25% (t = 1.7)	0.18

Figure 3. The 2008 carry crash, monthly returns July 2008 to March 2009.



From July 2008 to March 2009 the strategy lost **-20.5%**, as the funding currencies (yen and Swiss franc) rallied against the Australian and New Zealand dollars. Worst months: 2008-10 (-12.0%), 2007-08 (-7.8%), 2000-05 (-7.0%), 2009-01 (-6.8%), 2008-03 (-6.5%) and 2015-01 (-6.4%, the Swiss franc de-peg).

8.3 Strategy C: crypto funding carry (RQ4)

Figure 4. Growth of \$1, crypto funding carry (net, log scale), with the main events. The red band marks the Dogecoin and XRP rally of April–May 2021 while the strategy was short them.

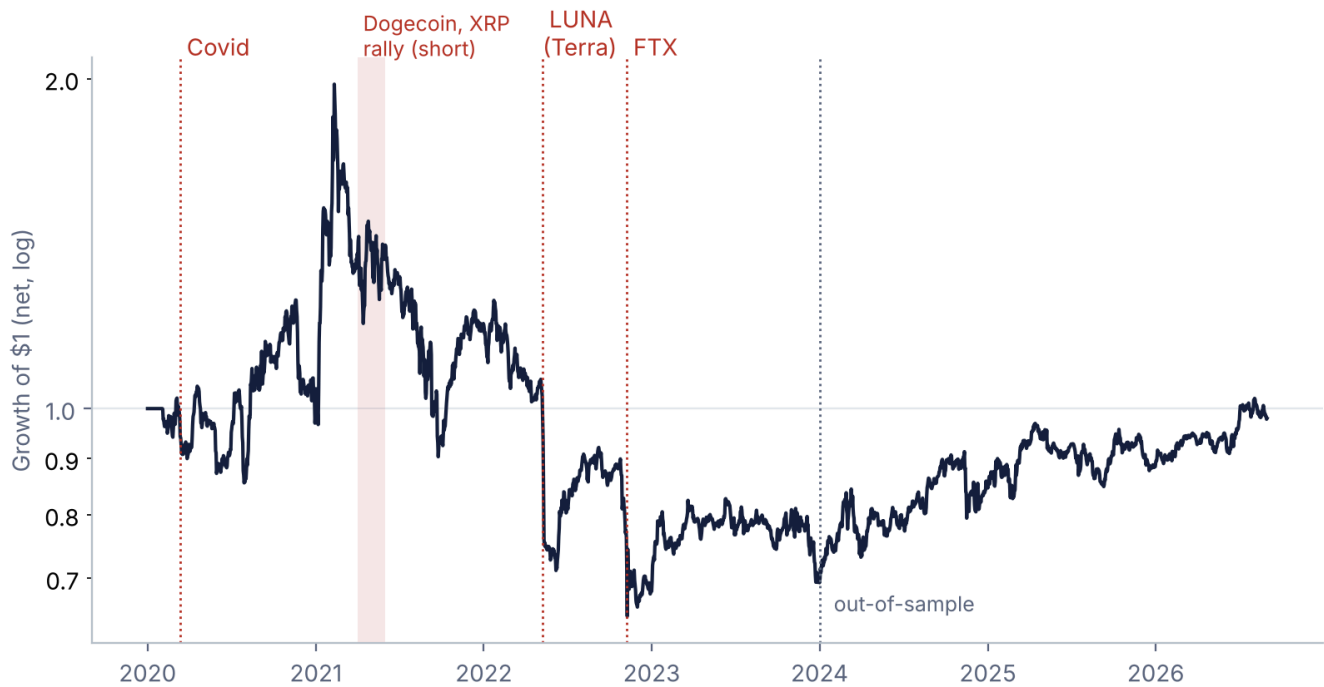


Table 5. Strategy C performance.

	Period	Ann. mean	Ann. vol	Sharpe	Max DD	Worst week	Turnover
Gross	Full 2020-02 to 2026-08	6.38%	29.87%	0.21	-66.0%	-27.2%	45x
Net	Full	4.13%	29.87%	0.14	-67.3%	-27.3%	45x
Net	In-sample 2020-2023	-2.52%	36.05%	-0.07	-67.3%	-27.3%	45x
Net	Out-of-sample 2024-2026	13.87%	17.21%	0.81	-13.3%	-9.2%	44x

Figure 5. Cumulative funding received vs price moves. The dotted green line marks the FTX crash (November 2022), when the strategy was long FTX Token and collected extreme funding.

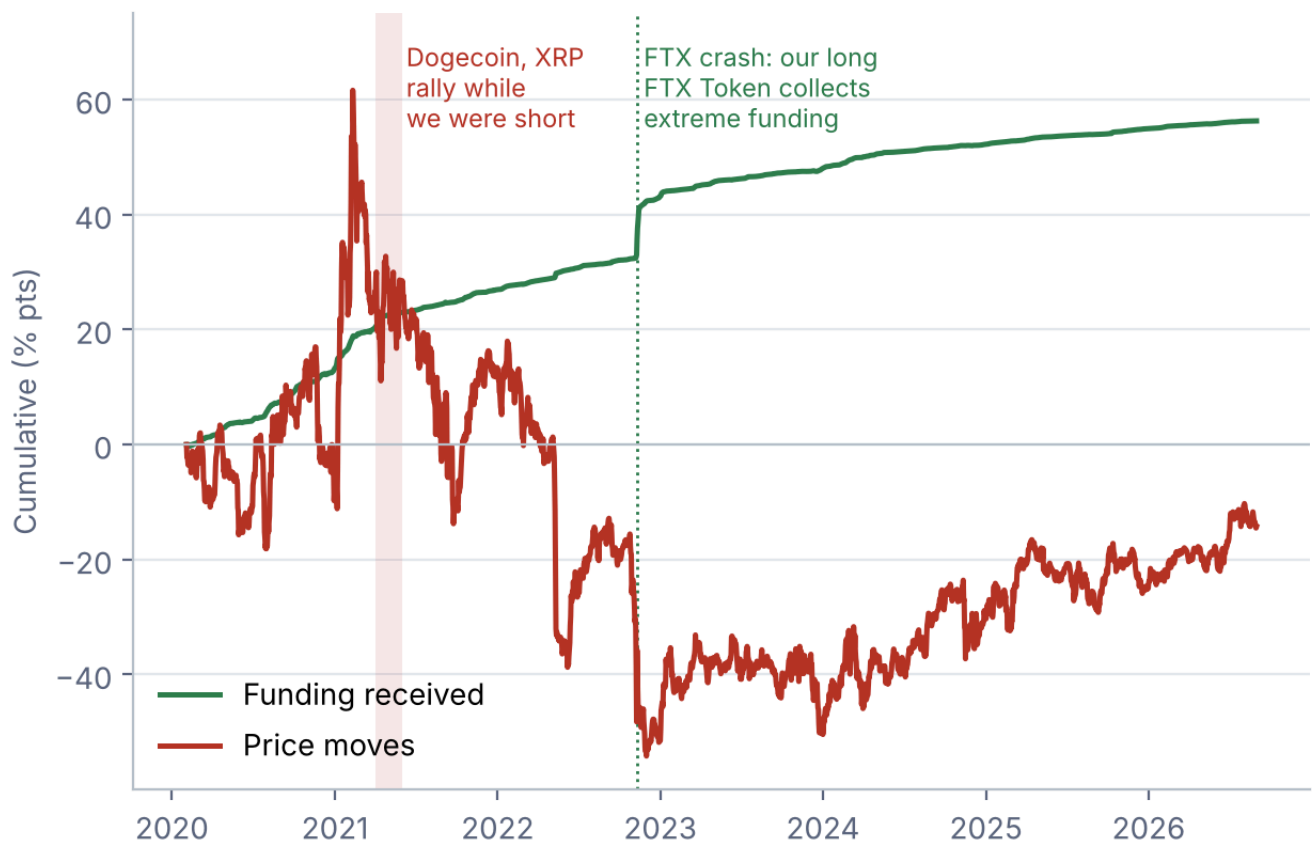


Table 6. Where the return comes from (% a year).

Period	Funding	Price	Costs	Net
Full	+8.55	-2.17	-2.25	+4.13
In-sample	+12.26	-12.51	-2.27	-2.52
Out-of-sample	+3.10	+12.99	-2.22	+13.87

Table 7. Robustness, lookback × rebalancing (net Sharpe).

Lookback	Rebalancing	Sharpe full	Sharpe out-of-sample
3 days	daily	0.68	0.41
3 days	weekly	0.49	1.29
7 days	daily	0.56	0.92
7 days	weekly (pre-set)	0.14	0.81
30 days	daily	0.28	0.83
30 days	weekly	0.07	0.42

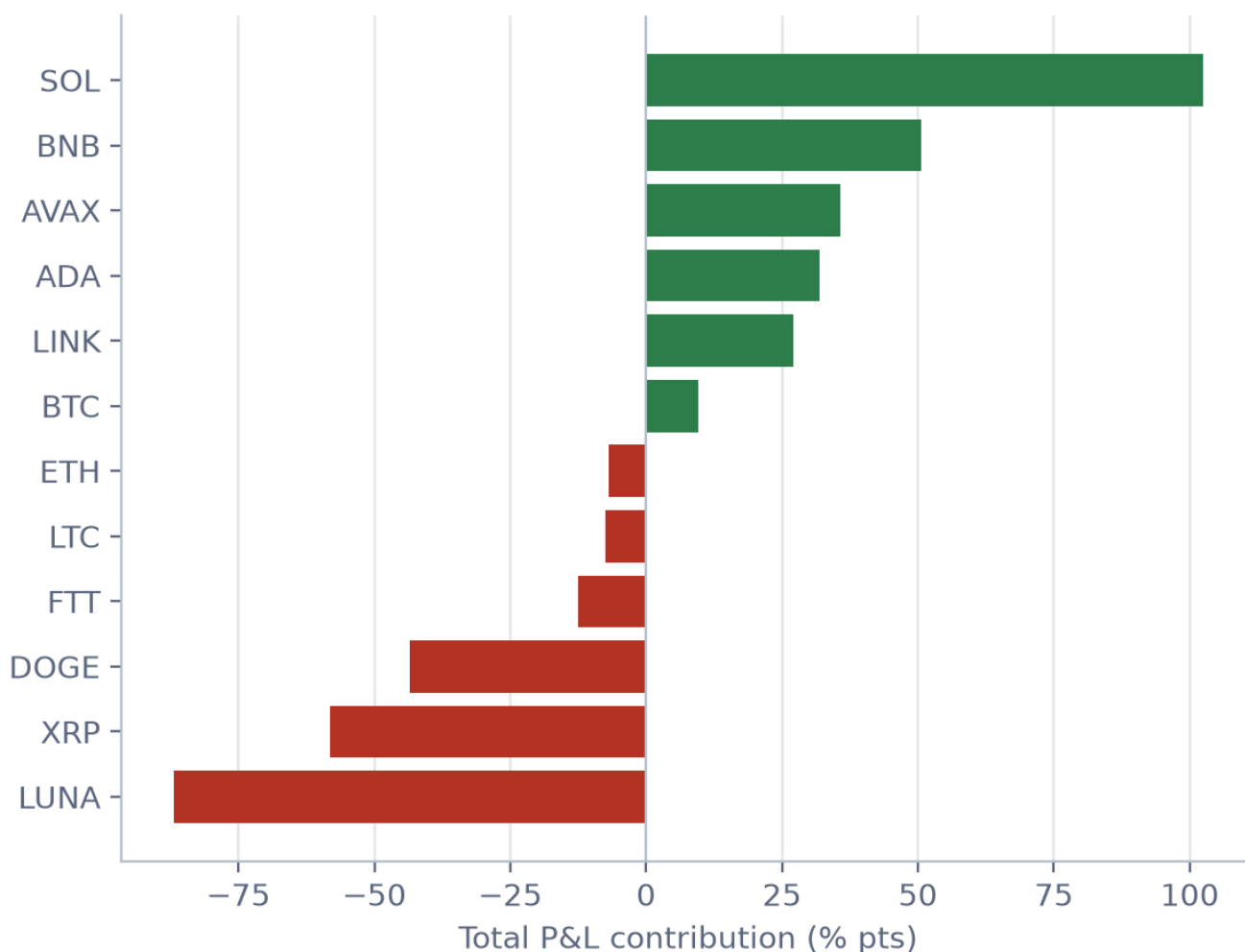
All 6 settings have a positive out-of-sample Sharpe (0.41 to 1.29), and the pre-set choice is in the bottom half, so we did not pick the best one after the fact.

The full diagnostic chart from the backtest script is [output/strategyC/strategyC_performance.png](#).

8.4 Strategy C: what drives the losses? (RQ5)

- **No market exposure.** Beta to daily BTC returns is -0.02 (Newey-West $t = -0.75$).
- **The worst week was Terra.** In the week ending 2022-05-15 the strategy lost -27.3% . It was *long* Terra: traders were shorting it, its 7-day funding had turned negative (about -20% a year), so the rule put it in the long leg. Other bad weeks: 2021-09-12 -12.3% , 2022-11-13 -11.6% (FTX), 2020-11-29 -11.0% , 2021-08-15 -10.0% .

Figure 6. Total P&L contribution by coin (% points).



Losses are coin-specific: Terra -87 , XRP -58 , Dogecoin -44 , FTX Token -13 ; winners Solana $+102$, BNB $+51$, Avalanche $+36$, Cardano $+32$, Chainlink $+27$.

8.5 Combined portfolio (RQ6)

Figure 7. A and C scaled to 10% volatility, and the 50/50 mix, 2021–2026.

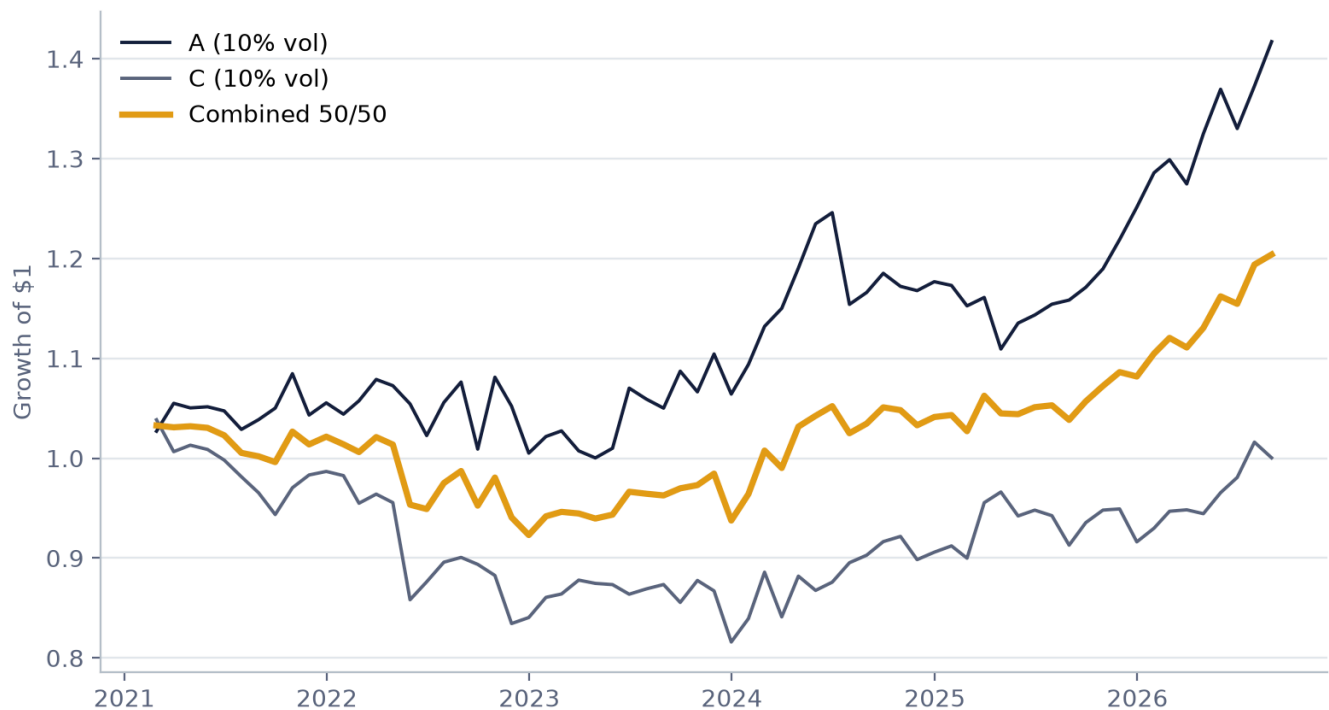


Table 8. Combined portfolio, 2021-02 to 2026-08 (67 months).

	Ann. mean	Ann. vol	Sharpe	Max DD	Worst month
A (10% vol)	6.70%	9.43%	0.71	-11.0%	-7.4%
C (10% vol)	0.44%	9.28%	0.05	-21.4%	-10.2%
Combined 50/50	3.57%	6.96%	0.51	-10.6%	-5.9%

The correlation between A and C is **0.11** (0.12 for unscaled monthly net returns, 78 months).

9. Hypotheses: verdict and evidence

Table 9. Verdict on each hypothesis.

Hypothesis	Verdict	Evidence
A1: G10 carry earns a premium	Supported	Net Sharpe 0.37; 0.82 correlation with the LRV factor
A2: crowding predicts weaker carry	Leaning yes	0.02% vs 0.30% next month; out-of-sample Sharpe 0.27 → 0.32; missed 2008
A3: carry loses in volatility spikes	Supported	Volatility beta -1.55 (t = -3.7); alpha not significant
C1: funding carry earns a premium	Mixed	Funding +8.5% a year, but net Sharpe 0.14; price moves dominate
C2: no BTC exposure; coin-specific crashes	Supported	BTC beta ≈ 0; Terra -87 points; worst week -27%

Hypothesis	Verdict	Evidence
FX and crypto carry diversify	Supported	Correlation 0.11; combined volatility 7.0% vs about 9%

10. Discussion

- **Strategy A (RQ1–RQ3).** G10 carry earns a modest premium (net Sharpe 0.37) and tracks the academic carry factor closely (correlation 0.82). Its losses line up with volatility shocks (beta -1.55 , $t = -3.7$), and the alpha is not significant once the dollar and volatility factors are included. This is the textbook picture: carry is pay for crash risk, not a free lunch.
- **Crowding (RQ2).** The evidence leans towards H1-A2 but is not decisive. After crowded months carry earns almost nothing and is more negatively skewed, and the filter raises the out-of-sample Sharpe from 0.27 to 0.32, with every setting in the robustness grid beating plain carry. But with only 33 crowded months the difference is imprecise, and the 2008 crash started from a non-crowded reading, so the filter did not avoid the largest drawdown.
- **Strategy C (RQ4–RQ5).** Crypto carry collects real funding income (+8.5% a year), but price moves dominate. In 2020–2023 the shorted, crowded-long coins such as Dogecoin and XRP kept rallying and wiped out the funding; since 2024 most of the gain came from price, not funding. The strategy has no BTC exposure, but its crashes are coin-specific and arise in the mirror image of FX: because it buys the coins with the most negative funding, it ends up long whatever traders short hardest. In 2022 that meant being long Terra and FTX Token into their collapse.
- **Diversification (RQ6).** FX and crypto carry are nearly uncorrelated (0.11), so the 50/50 mix has lower volatility (7.0%) and a smaller drawdown than either part. Because C was weak during the overlap, the mix does not beat A alone on Sharpe.
- **Implications.** In FX, crowding from positioning data is a cheap, modestly useful overlay. In crypto, sorting on funding alone exposes the portfolio to single-coin collapses; a better design would react faster than weekly (the robustness grid points that way), treat extreme negative funding as a sign of crowded shorts, and size positions by coin-level risk.

11. What is new

- We use **CFTC speculator positioning as a crowding filter** on G10 carry, and test it out of sample.
- We treat **crypto funding as both the crypto interest-rate gap and a crowding signal**, and test it on a universe that keeps the coins that died (Terra and FTX Token).
- We compare the two markets side by side with the same pre-registered design, and show that their crashes come from opposite sides of the crowd.

12. Limitations

- **Short crypto history.** About 6.5 years and only 12 coins; each leg usually holds 3 coins, so single-coin events dominate.
- **Few crowded FX months.** Only 33 crowded months, and the 2008 crash was not flagged.
- **Strategy C reverses** between in-sample (Sharpe -0.07) and out-of-sample (0.81). The out-of-sample period is short and benefited from price moves, so it is not evidence of a stable premium.
- **Signal horizon matters.** Shorter lookbacks and daily rebalancing do better even after costs; we report the pre-set case and did not switch after seeing the grid.
- **Data gaps.** EUR spot starts in 1999 (DEXGEUS not used), JPY rates in 2002 and CHF rates in 1999, so Strategy A starts in 1999-05. The LRV factors end in 2021-05, so the risk regressions cover 1999–2021 only.
- **Simple costs.** Costs are per-unit estimates; funding caps, borrow limits, slippage under stress and exchange counterparty risk are not modelled.

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Table 8	Combined portfolio	output/risk/combined_portfolio.csv
Table 9	Hypotheses: verdict and evidence	this README, section 9

Other outputs: LRV comparison [A_vs_LRV_HML.csv](#), worst months [A_worst_months.csv](#), BTC beta [C_beta_btc.csv](#), coin contributions [C_contribution_by_coin.csv](#), worst weeks [C_worst_weeks.csv](#), and the monthly and daily return series in each output/ folder.

14. Repository structure

```
code/          numbered scripts, run in order (see run_all.py)
  01_download_data_strategyA.py  FX, rates, CFTC -> data/clean/
  02_backtest_strategyA.py       Strategy A backtest -> output/strategyA/
  03_download_data_strategyC.py  Binance prices and funding -> data/clean/
  04_backtest_strategyC.py       Strategy C backtest -> output/strategyC/
  05_risk_analysis.py            LRV check, risk, combined -> output/risk/
  06_build_web.py                interactive page -> web/index.html
  07_build_slides.py             long deck and charts -> slides/presentation.pptx,
slides/img/
  08_build_slides_10min.py       10-minute deck with speaker notes ->
slides/presentation_10min.pptx
notebooks/      step-by-step Jupyter versions of the scripts (01-05)
data/raw/       raw downloads as received (FRED, CFTC, Binance, LRV file)
data/clean/     processed data snapshot used for all results (committed)
output/         result tables (CSV) and diagnostic figures
slides/         presentations, speaker notes and chart images (slides/img/)
web/           interactive HTML page
docs/          study guide (HTML and PDF, English and Persian)
submission/    the replication package zip submitted on Gradescope
paper/         journal-style paper (PDF, LaTeX source and figures)
```

15. How to reproduce

```
conda create -n carry python=3.12 -y    # or use your own environment
conda activate carry
pip install -r requirements.txt
SKIP_DOWNLOAD=1 python run_all.py        # all results, page and slides from the
committed data/clean snapshot
python run_all.py                        # full pipeline including downloads
```

- **SKIP_DOWNLOAD=1** reproduces every result from the committed data/clean/ snapshot, without network access.
- **FRED files must be downloaded by hand** for a full rebuild, because FRED often drops scripted requests. Save each series from fred.stlouisfed.org (Download → CSV) as data/raw/fred_<SERIES_ID>.csv, e.g. data/raw/fred_IR3TIB01USM156N.csv, or set FRED_API_KEY to use the official API.

- **code/05_risk_analysis.py** needs the course file **CurrencyPortfolios.xls** in data/raw/; the sections that use it are skipped if it is absent. A hand-saved data/raw/fred_VIXCLS.csv is used if present.
- **Notebooks.** notebooks/01–05 import the functions in code/, so their results match the scripts. They use data/clean/ by default; set DOWNLOAD = True in notebooks 01 and 03 to re-download.

```
python -m ipykernel install --user --name carry --display-name "Python  
(carry)"  
jupyter lab notebooks/
```